



A catfish Skin-inspired dual-antifouling epoxy thermoset as coatings for marine antifouling applications

Peizhan Cao^{a,b}, Shu Tian^{a,c,1}, Jinyue Dai^{a,c,*}, Jiayin Li^{a,b}, Guangming Lu^{a,c},
Xiaoqing Liu^{a,c,*}

^a Ningbo Institute of Materials Technology and Engineering, Chinese Academy of Sciences, Ningbo 315201, PR China

^b University of Chinese Academy of Sciences, Beijing 100049, PR China

^c Key Laboratory of Marine Materials and Related Technologies, Key Laboratory of Marine Materials and Protective Technologies of Zhejiang Province, Ningbo 315201, PR China

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ABSTRACT

Simultaneously endowing marine coatings with excellent intrinsic bactericidal and anti-adhesion properties is a subject of great practical significance and challenge. Inspired by the dual-antifouling effect of catfish, we developed epoxy thermoset as marine antifouling coatings with contact sterilization and anti-adhesion capabilities. Here, a dual-antifouling epoxy thermoset can be conveniently obtained by using bisphenol A diglycidyl ether (DGEBA) as the matrix resin, 1,3-diamino-2-hydroxypropane (DAHP) as the antibacterial curing agent, and polyethylene glycol dicarboxylate diglycidyl ether (PEGE) as the hydrophilic polymer. The results show that the obtained coating exhibits a higher bactericidal rate of more than 99.9 % against both *E. coli* and *S. aureus*. Epoxy resins cured using amines with a hydroxyl group at the β -position can generate carbon atoms with relatively positive charge characteristics in the resin network, which can strongly bind the phospholipid bilayer of bacterial cell membranes and exert bactericidal effects. This fact was first discovered and elucidated by us and provides a new paradigm for the easy design of antimicrobial resins. Moreover, the coating illustrates highly efficacious antifouling organisms adhesion, which, with a PEGE content above 30 %, has a comprehensive anti-protein and algae adhesion effect of over 98 %. The molecular dynamics (MD) simulations proved that this is attributed to a hydration layer formed by introduced PEG chains. For applications in complex marine environments, the coating maintains good mechanical properties with a tensile strength of 14.3 MPa, elongation at break of 68.4 %, adhesion, and impact strengths of 4.24 MPa and 60 kg-cm, respectively. This coating is highly suitable and promising to address the complex issue of marine biofouling.

1. Introduction

Biofouling is caused by the adhesion of proteins, bacteria, algae, and other marine organisms to the surface [1–4]. Forming biofouling presents a substantial risk to the marine industry, leading to considerable financial losses and environmental issues [5–8]. Using antifouling coating is an efficient and practical method to avoid marine biofouling. However, most antifouling agents (e.g., ZnPT, CuPT, DCOIT, diuron, etc.) pose environmental risks that threaten marine ecosystems [9]. Against this background, methods to achieve environmentally friendly and efficient prevention of marine biofouling are still being explored.

The use of efficient and environmentally friendly organic antifouling

coatings can provide reliable protection for a wide range of marine-related equipment. Epoxy resins, as versatile and designable thermosetting resins, are widely used in protective coatings for marine equipment because of their excellent mechanical properties, adhesion to various substrates, and considerable stability [10,11]. However, general-purpose bisphenol A epoxy resins do not kill microorganisms and are ineffective against the adhesion of a wide range of biofouling. In most cases, epoxy resin-based antifouling coatings rely on blending with antifouling agents to achieve antifouling effects [12,13]. Various natural, inorganic, and organic antimicrobial agents physically blended with resin systems have also been reported extensively [14–16]. For example, Ashraf M El Saeed et al. [15] reported the antifouling coating obtained

* Corresponding authors.

E-mail addresses: daijinyue@nimte.ac.cn (J. Dai), liuxq@nimte.ac.cn (X. Liu).

¹ Equal contribution to the first author.

by compounding cuprous oxide nanoparticles with epoxy resin as an antifouling agent and achieved a good antifouling effect, and Wu et al. [16] reported the antifouling epoxy resin by using natural *Artemisia annua* wastes as an effective biocide. However, the physical blending of antifouling agents with resins inevitably results in the exudation or release of antifouling agents, which may have a negative impact on the environment and inevitably lead to the deterioration or even disappearance of the antifouling performance of the coatings [17–19]. Covalent bonding of bactericidal structures in resins can avoid these problems [20,21]. Peng et al. [22] and Huang et al. [23] have carried out relevant studies in this regard, in which they covalently introduced imine and phenol-containing structures into the epoxy resin network and obtained intrinsically antimicrobial epoxy resins, respectively. However, resins with only a single bactericidal effect may shield the bactericidal surface from direct contact with various microorganisms and lead to coating failure once the surface is covered with proteins or fragments of dead bacteria and algae [24,25].

Therefore, in addition to bactericidal capability, antifouling coatings must also resist adhesion to various biofouling agents. Based on this viewpoint, antifouling coatings with bactericidal and anti-adhesive properties have also been reported [26–30]. Sun et al. [26] prepared epoxy coatings with both microbicidal and biofouling adhesion resistance based on stereochemistry on polymer surfaces and the release of microbicides and achieved good antifouling results. Similarly, Xie et al. [27] achieved synergistic antifouling with antimicrobial properties and prevention of fouling adhesion utilizing self-renewing surfaces and the release of natural antifouling agents. However, in most cases, the reported antimicrobial performance of the bifunctional coatings is still achieved by physically blending the antifouling agents, which again may face the possibility of antifouling agent bleed-off, leading to coating failure and environmental burden. The generation of microbicidal units in epoxy resin networks without the addition of antifouling agents while trying to prevent the adhesion of various biofouling is still a challenging task. Achieving such a functional design of epoxy resins can effectively avoid the drawbacks of blended antifouling agents and is expected to

result in a coating that is efficiently antifouled in the long term [31,32].

In our work, we report for the first time a facile method for the in situ generation of bactericidal structures in epoxy resin networks. With the help of this chemical structure, we designed a dual-antifouling epoxy coating that mimics the skin of a catfish (Fig. 1a). We mixed PEGE with DGEBA and chose an amine with β -hydroxyl groups as the curing agent. The β -amino alcohol structure of the curing agent can produce positively charged carbon atoms through the induced effect of the oxygen atom on the hydroxyl group and the nitrogen atom at the β -position, which can act as a bactericidal agent. This bactericidal structure is functionally equivalent to the antimicrobial peptide with bactericidal properties naturally present on the skin surface of catfish. At the same time, the PEG chains introduced in this way are capable of forming a hydration layer to resist the adhesion of dirt organisms, which is comparable to the mucus that acts as a natural hydration layer present on the surface of catfish skin (Fig. 1b) [33,34]. With this design, we covalently bonded special bactericidal structures and PEG chains into an epoxy resin network to achieve an antifouling coating that mimics the functionality of a catfish's skin, which kills bacteria that come into direct contact with it while resisting the adhesion of various biofouling agents. The design achieves a highly effective antifouling effect and also avoids the need to rely on mixed antifouling agents. This offers a practical solution to the demand of the marine sector for eco-friendly and effective antifouling coatings.

2. Experimental section

2.1. Materials

The Bisphenol A diglycidyl ether (DGEBA, Epoxy value = 0.51) was purchased from Baling Co., SINOPEC. The Polyethylene glycol diglycidyl ether (PEGE, Epoxy value = 0.38), 1,3-diaminopropane (DAP, 98 %), 1,3-diamino-2-hydroxy propane (DAHP, 97 %) were purchased from Macklin Biochemical Co., Ltd, Shanghai, China. All chemicals and solvents were used without further purification.

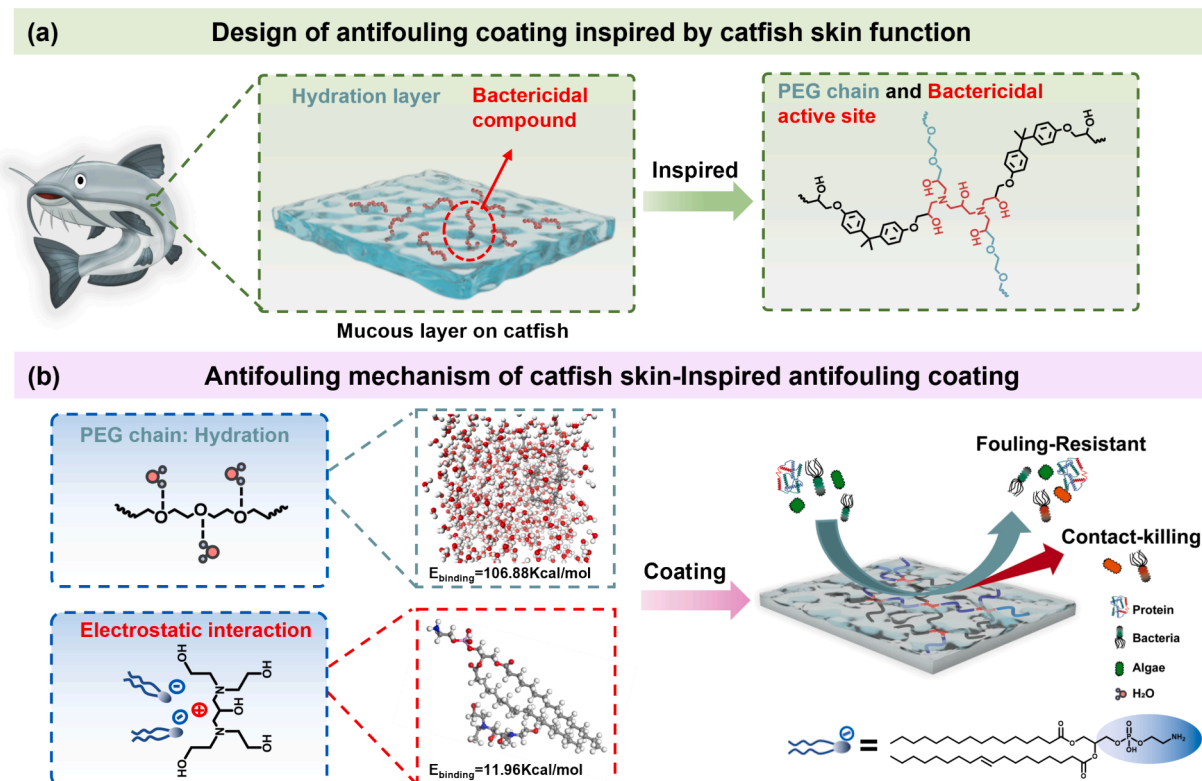


Fig. 1. Schematic design concept for antifouling coatings.

2.2. Preparation of coatings

The design concept of the coating and the structure of the designed coating are shown in Fig. 1. To obtain coatings that are bactericidal and have anti-adhesive properties, DEGBA and PEGE were mixed in a set ratio, with DAHP as the curing agent. The system was stirred to disperse homogeneously and placed in a beaker at room temperature under a vacuum to remove air bubbles. The mixed resin was coated on stainless steel with a 400 μm thickness applicator. Then, the epoxy resin was cured at 50 $^{\circ}\text{C}$, 70 $^{\circ}\text{C}$, and 90 $^{\circ}\text{C}$ for three hours, respectively, to obtain epoxy resin films with multiple β -amino alcohol characteristic structures. The same method and curing process were used to cure DEGBA with DAP to obtain theoretically nonsterile samples with no anti-adhesive ability. The same method of mixing DAP and PEGE and curing at 70 $^{\circ}\text{C}$, 90 $^{\circ}\text{C}$, and 110 $^{\circ}\text{C}$ for 3 h, respectively, to obtain the theoretical non-sterilizing but anti-adhesive ability of the samples. The feed compositions and schematic diagram of the preparation of the coatings are listed in Table S1 and Fig. S1 in the supporting information.

2.3. Characterizations

The curing activity of the epoxy resins was examined by differential scanning calorimetry (DSC, Mettler-Toledo DSC, Switzerland). Tests were performed in a nitrogen atmosphere (flow rate of 50 mL/min) at a heating rate of 5 $^{\circ}\text{C}/\text{min}$ and a temperature range of 25 $^{\circ}\text{C}$ to 150 $^{\circ}\text{C}$. The structure of the cured resins was measured using a micro-infrared spectrometer (Agilent, USA) in attenuated total reflection mode with a spectral recording range of 400–4000 cm^{-1} , 32 scans, and a resolution of 4 cm^{-1} . The transparency of the cured resins was obtained by the transmission mode of a UV-Vis near-infrared spectrophotometer (LAMBD A); the sample was approximately 400 μm thick, and the spectral recording range was 200–850 nm. The morphology of the coatings was observed by field emission scanning electron microscopy (SEM, S4800, Hitachi, Japan). The thermal stability of the cured resins was obtained by thermogravimetric analysis (TGA, TG209F1, Germany), and the tests were performed in a nitrogen atmosphere with a heating rate of 10 $^{\circ}\text{C}/\text{min}$ and a temperature range of 25 $^{\circ}\text{C}$ to 800 $^{\circ}\text{C}$. The dynamic mechanical properties, as well as the glass transition temperature (T_g) of the cured resins, were measured by a dynamic thermomechanical analyzer (DMA, DMAQ800, USA) in a single cantilever mode with a sample size of 30 mm $\times$ 8 mm $\times$ 2 mm, and a temperature range of -50 $^{\circ}\text{C}$ to 200 $^{\circ}\text{C}$, with a heating rate of 5 $^{\circ}\text{C}/\text{min}$, a frequency of 1 Hz, and an amplitude of 10 μm . The tensile properties of the cured resins were tested according to GB/T 1040.2–2006 for testing, the use of dumbbell-shaped sample strips, marking distance of 30 mm, tensile rate of 5 mm/min $^{-1}$, each curing system tensile experiment repeated five times to take the average value. The pencil hardness of the coatings was tested by a pencil hardness tester (QHQA, Ltd., Japan) according to ASTM D3363-00; the adhesion was tested by an adhesion tester (PosiTest AT-M, USA) according to GB/T 5210; the impact resistance of coating films was tested through a Paint film impactor (QCJ50, China) according to GB/T 1732. A contact angle meter (Data Physics, Germany) was used to detect the contact angle and the change in contact angle over time. A drop of water (3 μL) was applied to the coated surface at room temperature, and the contact angle was recorded every 30 s. A total of 20 sets of data were reported. The surface roughness of the coating is determined by a 3D optical profilometer (UP-Lambda, USA). Determination of salt water resistance of marine coatings is carried out according to GB/T 10834–89.

2.4. Antifouling property

2.4.1. Protein adsorption tests

BSA-FITC was dissolved in PBS to obtain a 2 mg/mL BSA-FITC solution. The coatings to be tested were immersed in the configured BSA-FITC solution for 4 h and protected from light. Then, the coatings were

removed from the solution and rinsed thoroughly with PBS solution. The treated coatings were observed and photographed using an inverted fluorescence microscope. Three replicate experiments were performed for each coating. The relative adhesion rate was counted by ImageJ software and the mean and standard deviation were calculated.

2.4.2. Antibacterial tests

The inhibitory effect of the coatings on *Escherichia coli* (ATCC8739, Gram-negative) and *Staphylococcus aureus* (ATCC6538, Gram-positive) was verified according to the standard for testing antimicrobial activity on plastic surfaces (ISO 22196:2007, IDT), respectively, and the bacterial concentration in the experiments was 10^5 CFU/mL. Before the tests were carried out, the solution used was sterilized in an autoclave for 15 min. The surface of the samples was wiped with alcohol, and the experiment was carried out after the alcohol had evaporated completely. The bacterial solution on each sample was incubated in three separate agar plates. And the number of colonies grown on the agar plates was counted by an automatic colony counter (Scan300, Interscience, France). The antimicrobial rate (A.R.) was calculated as follows:

$$\text{A.R.} = \frac{A_0 - A_t}{A_0} \times 100\%$$

A_0 is the number of microbial colonies on the iron sheet and represents the number of microbial colonies in the coated condition.

E. coli and *S. aureus* were selected to evaluate the resistance of the coatings to bacterial adhesion behavior. The coatings were cut into 10 mm $\times$ 10 mm squares, placed in 24-well plates, inoculated with 100 μL of bacterial solution at a concentration of 10^8 , and incubated at 37 $^{\circ}\text{C}$ for twelve hours. Then, bacteria that did not adhere to the surface of the coatings were removed by rinsing with PBS solution. Bacteria adhering to the coated surface were stained using a calcein AM/PI kit and observed by inverted fluorescence microscopy. Three replicate experiments were performed for each coating and the photographs obtained were analyzed using ImageJ software. The mean and standard deviation of the relative attachment rates of the two bacteria were calculated. Meanwhile, the samples with adhered bacteria were fixed by glutaraldehyde (2.5 %) and washed with graded concentrations of ethanol, and SEM observed the morphology of dead bacteria.

2.4.3. Alga resistance tests

Chlorella vulgaris and *Phaeodactylum tricornutum* were selected to study the attachment of algae to the coated surface. The coatings were cut into 10 mm $\times$ 10 mm squares, placed in 24-well plates, inoculated with a suspension of algae at an initial concentration of 10^8 cells per milliliter, and incubated for 24 h at room temperature. Then, the algae that did not adhere to the coating surface were removed by rinsing it with artificial seawater. Three replicate experiments were performed for each coating. The rinsed coatings were observed using an inverted fluorescence microscope, and the resulting photographs were analyzed using ImageJ software. The mean and standard deviation of the relative attachment rates of the two algae were calculated.

2.5. Calculation details

All quantum chemistry calculations were done with the Materials Studio software. The binding energy of the PEGE repeating unit, DEGBA repeating unit, and water molecules in the coatings was simulated by molecular dynamics (MD) using the Forcite module. In a 3D simulation box, a strip of repeating units, as well as 500 water molecules, were placed for simulation. The interaction between repeating units and H_2O was supposed to be in a three-dimensional simulation box using a COMPASS II force field with the periodic boundary conditions at 298 K. An NVT canonical systematic with 500 ps simulation time and one fs time step was implemented in all simulations. The binding energy of DAHP-Derivative and POPE, as well as DAP-Derivative and POPE, were

calculated using the same method as above [35,36]. The binding energy of DAP-Derivative with POPE and DAHP-Derivative with POPE was also simulated using the Forcite module. The following equation calculated the binding energy of the system:

$$E_{\text{binding}} = -E_{\text{intercat}} = -(E_{\text{tot}} - E_a - E_b)$$

where E_{intercat} represents the interaction energy of a and b, and E_{tot} , E_a , and E_b represent the total energy of the a, b interacting system, and the energy of the system in which a and b are alone [37].

The atomic charge and electrostatic potential maps of DAP-Derivative and DAHP-Derivative in Fig. 7a are calculated using the DMol3 module, the BLYP generalization of the GGA generalization. The geometrical optimization of the above two structures was performed first, and then the energy states of the systems were calculated to obtain data on electron density, electrostatics, and population [38].

3. Results and discussion

3.1. Curing behaviors

The curing behavior of all the resins in Fig. 2a is similar, showing a single curing peak reflecting the reaction of the epoxy groups with the active hydrogen on the amine group. The exothermic peaks become lower and wider with the increase of PEGE content, which proves that the reactivity of PEGE is lower. In addition, DAHP has higher reactivity than DAP due to the hydrogen bonding between the hydroxyl group in the β -hydroxylamine and the oxygen on the epoxy [39]. Fig. 2b shows the infrared spectra of the cured coatings. Firstly, compared to the

spectra of uncured DGEBA and PEGE (Fig. S2), the disappearance of the epoxy peak at approximately 910 cm^{-1} and the appearance of hydroxyl peaks at 3500 cm^{-1} proved that the epoxy groups were involved in the curing reaction, and cured epoxy resin was obtained [40]. Besides, the enhanced peak of the C-O stretching near 1090 cm^{-1} and the weakened peak of the benzene ring backbone from 1600 cm^{-1} to 1450 cm^{-1} proved the introduction of the PEG chain [41]. In addition, it was also discovered that the resulting resins exhibited good transparency, and Fig. S3 in the supporting information contained the presentation of the light transmission data, which had been characterized using a UV-visible near-infrared spectrophotometer.

3.2. The fracture surface of the coatings

SEM images of the brittle fracture of the resin sections under liquid nitrogen freezing conditions are shown in Fig. 2c. When the PEG fraction is less, the smooth and flat sections show the typical characteristics of brittle fracture. In contrast, the ductile fracture can be observed in the section of EP-PEG without bisphenol A fraction [42]. When the weight fraction of PEGE is less than 40 %, all resin components have good compatibility, and the system presents a homogeneous phase. When the PEGE weight fraction is 50 %, the DGEBA and PEG components are no longer completely soluble in each other, and the undissolved PEG chain segments are uniformly dispersed in the polymer in an agglomerated state, which can provide a significant toughening effect.

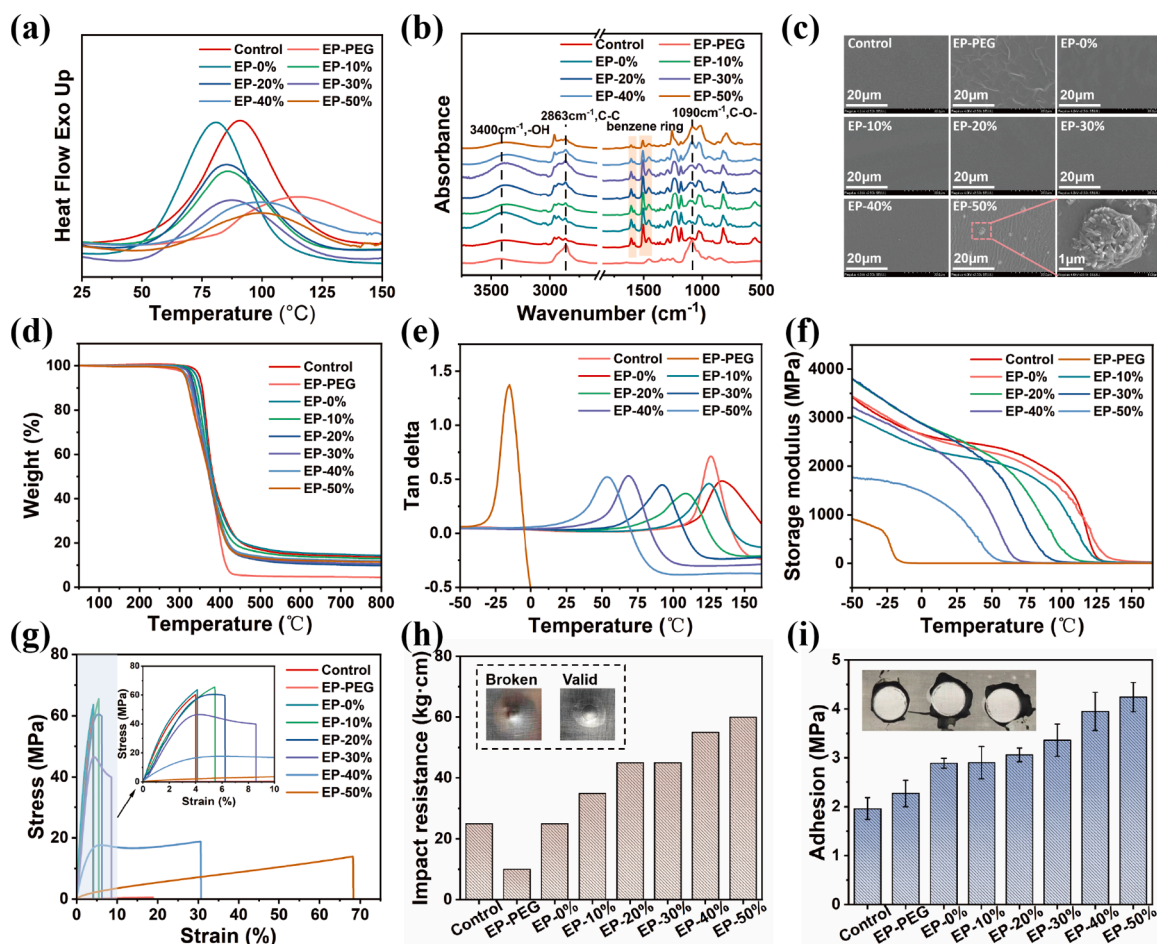


Fig. 2. (a) DSC curves, (b) The FT-IR spectrum of epoxy resins, (c) SEM images of the cross-section of resins, (d) TGA, (e) and (f) DMA of resins, (g) Tensile stress-strain curves of resins, (h) Impact resistance and (i) adhesion of coatings.

3.3. Thermal and mechanical properties of resins

The thermal stability of the resins in nitrogen was investigated by thermogravimetric analysis, and Fig. 2d shows the thermal weight loss curves of the resins. All cured resins have similar thermal weight loss behavior: they remain stable below 300 °C and $T_{d,10\%}$ (10 % weight loss temperature) all are higher than 320 °C, exhibiting good thermal stability. Fig. S4 shows the DTG curves of the cured resins, with the introduction of PEG chain segments, two extreme values of heat loss rate occur, which is attributed to the fact that the long chain PEG repeat units are less thermally stable and begin to decompose at lower temperatures, while the crosslinking point portion is highly thermally stable and has a greater heat loss rate at higher temperatures [43].

DMA evaluated the dynamic mechanical properties of the cured resins, and the results are shown in Fig. 2e and f. The storage modulus of EP-PEG is around 2.4 MPa at 25°C. However, when the mass fraction of DGEBA is raised to 50 %, the storage modulus of the cured resin surpasses 1000 MPa. The introduction of bisphenol A chains dramatically improves the mechanical properties of the material and makes it valuable for applications. Fig. 2e shows the loss factors of different cured resins, and the maximum value corresponds to the glass transition temperature (T_g) of the resin. The T_g of EP-PEG is about -15.2 °C, while the T_g of EP-50 % is about 54.2 °C, and this temperature is sufficient for the application of the material at room temperature. The T_g is similarly improved with increased DGEBA content and up to 134.8 °C for EP-0 %. The results of the tensile tests at 25 °C are shown in Fig. 2g, where EP-PEG exhibits a tensile strength of about 0.56 MPa and an elongation at break of 18.9 %, showing fragile mechanical properties. In contrast, Control exhibits a tensile strength of 64.8 MPa and an elongation at break of 4.8 %, with high strength but poor toughness. The introduction of PEG chain segment restriction into the DGEBA increased the toughness of the material, and EP-50 % exhibited a tensile strength of 14.3 MPa and an elongation at break of 68.4 %, which optimized the overall performance of the resins. The introduction of PEG chain segments into the DGEBA balanced the strength and toughness of the material. Between all the samples, EP30%, EP-40 % and EP-50 % showed a good balance of strength and toughness, avoiding over-brittleness or under-strength, and having a very good potential to be used as a coating.

Applying the resin in the form of a coating, hardness, impact strength and adhesion are performance indicators that must be considered [11]. Various data related to the coatings are shown in Fig. 2h, i and Table 1. Control and EP-0 % without PEG chain showed a high pencil hardness of 7H. In contrast, the hardness of the coatings gradually decreased with the increase of the PEG chain, the pencil hardness of EP-50 % was 3H, and the pencil hardness of EP-PEG without bisphenol A chain was only B. This shows the poor mechanical properties of pure PEG and that DGEBA is necessary for the hardness of the coating. The trends of adhesion and impact strength in Fig. 2h and i were similar. They increased with the increase in the content of the PEG chain, where the adhesion of EP-50 % was close to 4.5 MPa, and the impact strength was 60 kg·cm. This result was attributed to the fact that the flexible PEG chain segments acted as a toughening effect to reduce cohesive energy and improve the impact strength of the coatings. The adhesion is related

to the hydroxyl group and flexibility of the surface, and the introduction of PEG increases the toughness and, therefore, improves the adhesion similarly. The increase in PEG content has a negative effect on hardness, but a certain amount of PEG can improve adhesion and impact strength. In addition to the basic mechanical properties of the coating, the PEG content for optimum performance should be considered in light of its marine antifouling capability. In addition, considering that marine antifouling coatings need to meet the demands of long-term service under seawater, we investigated the saltwater resistance test of the coating, and the relevant results are shown in Fig. S5.

3.4. Antifouling property test

3.4.1. Anti-protein adhesion of the coatings

Efficient antifouling coatings must effectively resist the adhesion of proteins. Organic macromolecules, particularly proteins and polysaccharides, typically adsorb within minutes of immersion in the marine environment, forming a conditioning film that facilitates the adhesion of marine microorganisms, such as bacteria and algae [44]. Designing coatings that contain PEG structures is a standard method for achieving resistance to protein adhesion. BSA-FITC was selected to evaluate the anti-protein properties of the coatings. Fig. 3a demonstrates that the Control group, which lacked PEG chain segments, exhibited attachment by BSA-FITC. In contrast, EP-PEG only displayed a weak fluorescent signal, with a relative adhesion rate of just 2.79 %, indicating a 97.21 % reduction in protein adhesion. These results demonstrate the effectiveness of PEG chain segments in resisting protein adhesion. However, observing fluorescence signals in all coatings containing the multiple β -amino alcohols structure was difficult. This is understandable for coatings with high PEG content, which can significantly resist protein adhesion by forming a hydrated layer. However, this phenomenon is anomalous for EP-0 % without any PEG. We attribute this phenomenon to the fluorescent properties of BSA, which derives its fluorescent signals from amino acids with fluorescent properties in its structure. The dominant tryptophan produces a fluorescence burst when exposed to a certain degree of polarity [45]. In our resin network, the strong induced effect of nitrogen and the hydroxyl group directly connected to the central carbon atom leads to the presence of strong polar covalent bonds, and it is reasonable to assume that the strong polar structure due to this induced effect also leads to the fluorescence quenching reaction. Therefore, it is also difficult to observe the fluorescence signal of BSA on the EP-0 % sample.

3.4.2. Antibacterial properties

In the marine environment, the attachment of bacteria to metals or other materials to form complex communities can cause many industrial problems, such as biocorrosion, and provide conditions for the attachment of various large marine fouling organisms [44]. Therefore, antimicrobial capacity is an essential property of marine antifouling coatings. The antimicrobial characterization structure of multiple β -amino alcohols was introduced into the curing resins to obtain a series of antifouling coatings, which resulted in the contact killing of bacteria. The characteristic structure that exerts the bactericidal effect is covalently linked, so there is no release of toxic substances into the environment, and harm to non-targets in the marine environment can be avoided with environmentally friendly properties [46]. The bactericidal ability of the coatings against *E. coli* and *S. aureus* was evaluated by plate counting method, and the relevant results are displayed in Fig. 4a and b. The number of bacteria on iron sheets, Control, and EP-PEG was comparable, indicating that the coatings obtained by curing DGEBA as well as PEGE with DAP as the curing agent had little or no bactericidal effect against *E. coli* and *S. aureus*. On the other hand, the coatings with β -amino alcohol characteristic structure obtained by using DAHP as the curing agent did not cultivate any bacteria, and the bactericidal rate reached 100 %. Among the samples used in the experiments, Control and EP-0 % only used different curing agents, yet they demonstrated a

Table 1
Thermal and mechanical properties of different resins.

Coating sample	T_g (°C)	$T_{d,10\%}$ (°C)	Char yield (%)	E (MPa) at 25°C	Hardness
Control	126.6	354.9	13.7	2520.1	7H
EP-PEG	-15.2	327.5	4.5	2.4	B
EP-0 %	134.8	350.1	14.3	2432.0	7H
EP-10 %	125.1	341.7	12.9	2219.2	6H
EP-20 %	108.9	336.1	9.8	2556.7	5H
EP-30 %	92.5	332.5	10.5	2490.2	5H
EP-40 %	68.9	327.8	11.8	1989.1	4H
EP-50 %	54.2	322.2	11.4	1020.5	3H

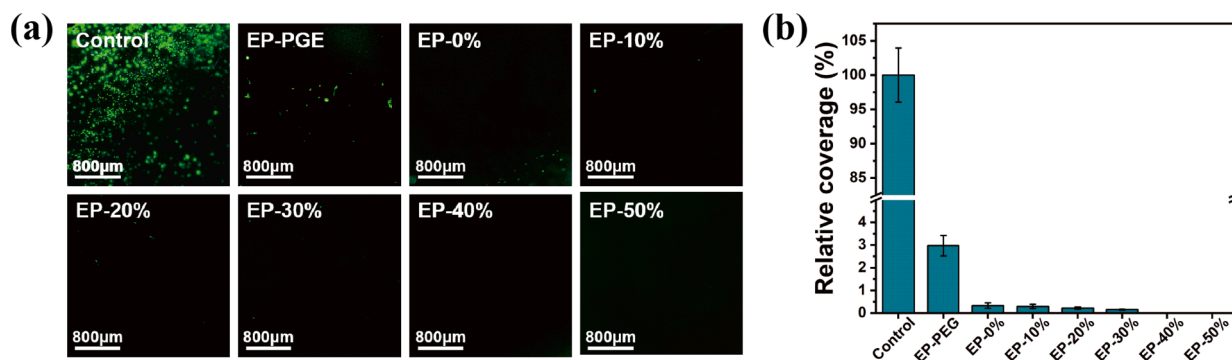


Fig. 3. (a) Inverted fluorescence microscope images of BSA-FITC attachment conditions, (b) BSA-FITC relative coverage area of the coatings.

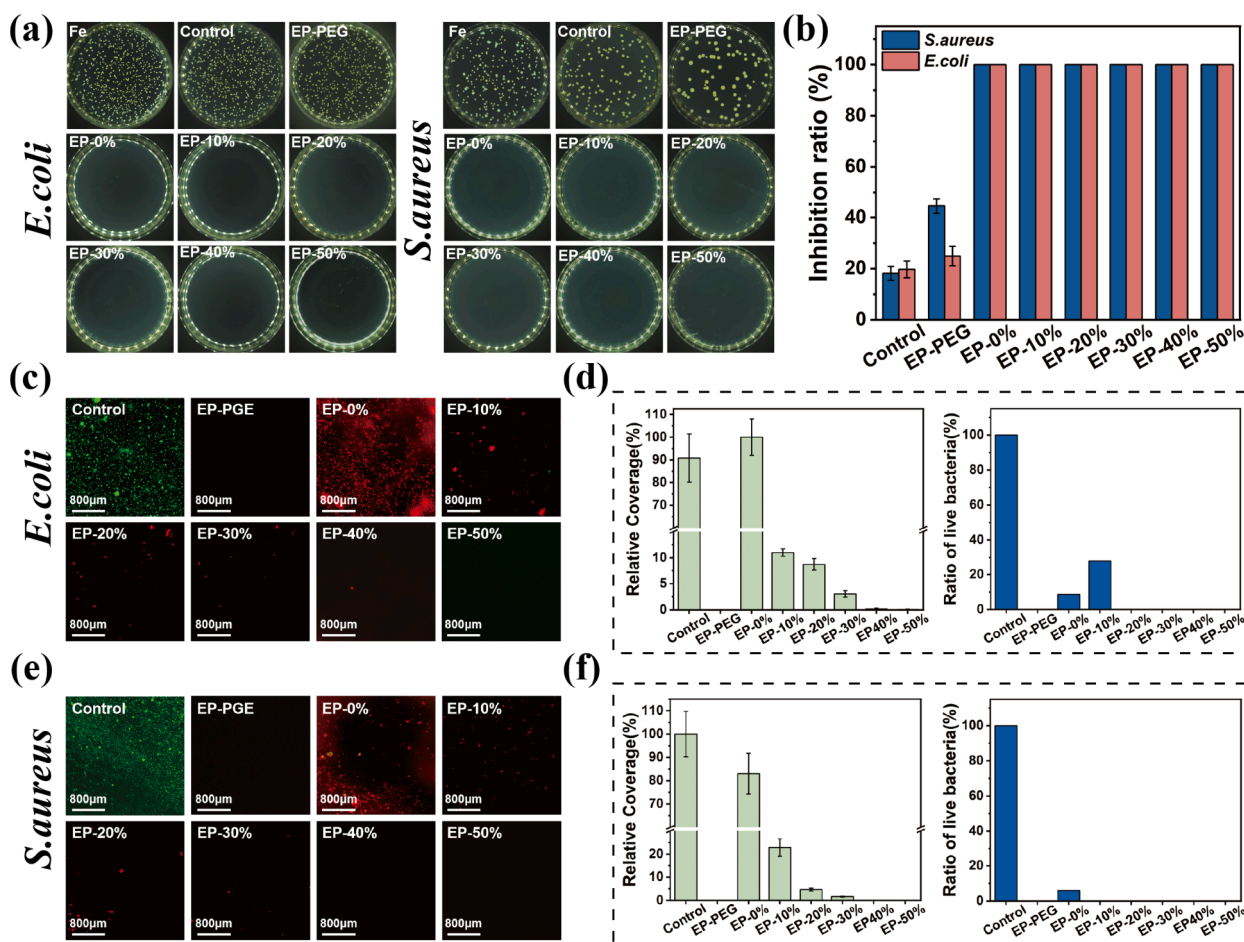


Fig. 4. (a) Digital picture of colonies, (b) Antibacterial effect, (c) The bacteria adhesion of *E. coli* on the coatings, (d) *E. coli* relative coverage area of the coatings and percentage of live *E. coli* on the coating, (e) The bacteria adhesion of *S. aureus* on the coatings, (f) *S. aureus* relative coverage area of the coatings and percentage of live *S. aureus* on the coating.

significant difference in antimicrobial capacity. Thus, it can be demonstrated that the antimicrobial capacity of the resin is obtained by curing the epoxy resin with β -amino alcohol. In addition, due to the higher curing activity of curing agents containing hydroxyl groups in the β position of the amine group, the EP-0 % system cured more thoroughly than the control [47], yet it was more bactericidal, which excludes the possibility that residual primary and tertiary amines contribute to the antimicrobial capacity. On the other hand, the FT-IR spectra showed the disappearance of the epoxy and amine peaks after curing (Fig. 2a and Fig. S2), which is again favorable to exclude that the antimicrobial capacity of the resin is due to residual amine groups. In addition, we

verified the antimicrobial effect of some of the coatings after immersion in artificial seawater. The coatings were still able to maintain close to 100 % antimicrobial resistance against *E. coli* and *S. aureus* after ten days of immersion, indicating that the bactericidal properties of the coatings are long-lasting (Fig. S6). This is due to the fact that the effective antimicrobial structure does not leach out as a result of immersion in seawater, and therefore the bactericidal ability of the coatings does not disappear.

Fig. 4c and e display the fluorescence images of *E. coli* and *S. aureus* adhered to the coatings. The ability to resist bacterial adhesion of the coatings was significantly improved by introducing PEG chain segments.

As a result, the percentage of *E. coli* and *S. aureus* adhering to the coating was reduced considerably. We counted the relative adhesion rates of other samples using Control or EP-0 % of the bacteria with the highest number of adherents as a reference, and the results are shown in Fig. 4d and f. When the content of PEGE reached 40 %, the bacteria almost no longer adhered to the surface of the coating. It is important to note that the bacteria adhering to the control surfaces were all viable. However, the percentage of viable bacteria adhering to the surfaces of the coatings with the characteristic structure of β -amino alcohols was meager. Evidence from fluorescence microscopy again demonstrates that these coatings can kill bacteria on contact and also demonstrates the need to introduce anti-adhesion features into the coating. Only bactericidal coatings can kill bacteria in direct contact with their surfaces. However, they are not able to prevent the adhesion of dead bacteria. The introduction of PEG chains in biocidal coatings greatly avoids the adhesion of both live and dead bacteria, which facilitates the long-term effectiveness of the biocidal surface and, in synergy with it, achieves a longer-term and more efficient antifouling effect.

3.4.3. Alga resistance of the coatings

Algal attachment occurs at the early stage of marine fouling formation, providing nutrient conditions and suitable sites for the attachment of large marine fouling [44]. Therefore, representative *Phaeodactylum tricornutum* and *Chlorella vulgaris* were chosen to study the resistance of the coatings to algal attachment. As shown in Fig. 5a and c, the fluorescence microscope photographs of *Phaeodactylum tricornutum* and *Chlorella vulgaris* adhering to the coatings are shown, respectively. The adhesion of the two algae on different coatings had a similar pattern. The adhesion rate of the algae decreased significantly with the increase in the content of the PEG chain. Introducing a 30 % mass fraction of the PEG component resulted in a 99.5 % decrease in the adhesion rate of

both algae relative to Control. In addition, it is worth noting that even though the control and EP-0 % do not contain PEG chains, there is still a difference in the adhesion behavior of algae on the surfaces of the two. On the one hand, the relative adhesion rate of algae to EP-0 % was still significantly lower than that of the control. On the other hand, compared with Control, the distribution of fluorescence signals of EP-0 % showed the characteristics of local agglomeration and overall dispersion. This phenomenon is due to the microbiocidal effect of EP-0 %, which kills algae in direct contact with it, thus preventing the algae from adhering uniformly to the coating. However, partial adherence of the dead algae to the coating surface shielded subsequent algae from direct contact with the coating, and the algae thus spread and grew centered on this and formed a locally clustered fluorescence signal [48]. Therefore, it is difficult for a coating that only kills microorganisms to avoid the adhesion of dead microorganisms and the resulting adhesion and proliferation of subsequent microorganisms. The introduction of PEG chains into the bactericidal coating was necessary to achieve resistance to the adhesion of biofouling in various states. The resulting coating with dual-antifouling capability shows a rather low relative algal adhesion rate, proving the efficiency of its bifunctional antifouling properties.

3.4.4. The discussion of anti-adhesion and sterilization mechanism

Coatings are designed with the idea that all ingredients can fulfill their unique purpose. In this work, covalent cross-linking homogeneously immobilized the PEG chains in the resin matrix. The above work demonstrated that introducing PEG chains significantly improved the resistance of the coating to adhesion. The calculation results proved that compared with the DGEBA repeating units, the PEG repeating units could bind water molecules more strongly. As shown in Fig. 6a, the PEG repeating units were more tightly bound to 500 water molecules, and

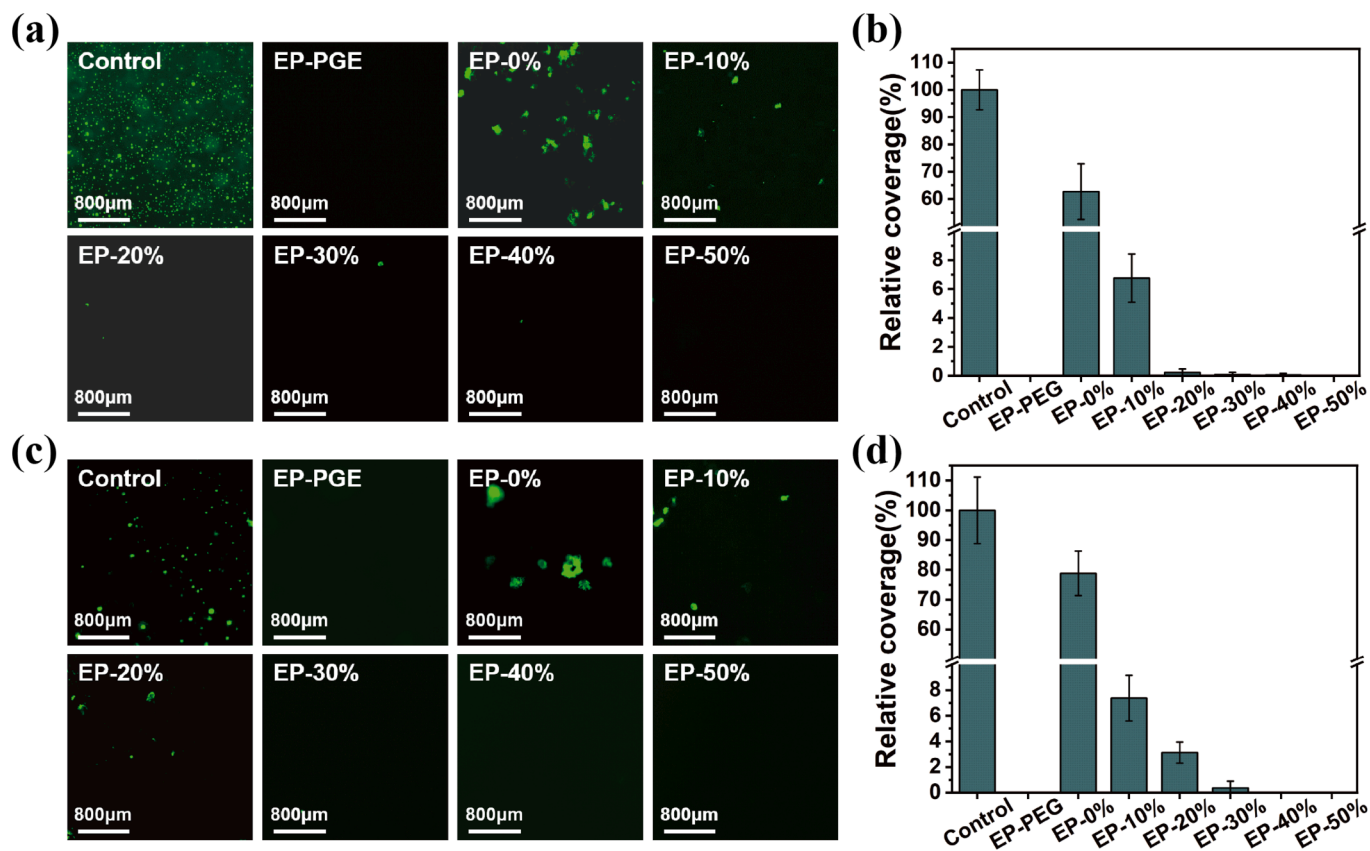


Fig. 5. (a) Inverted fluorescence microscope images of *Chlorella vulgaris* attachment conditions, (b) *Chlorella vulgaris* relative coverage area of the coatings, (c) Inverted fluorescence microscope images of *Phaeodactylum tricornutum* attachment conditions, and (d) *Phaeodactylum tricornutum* relative coverage area of the coatings.

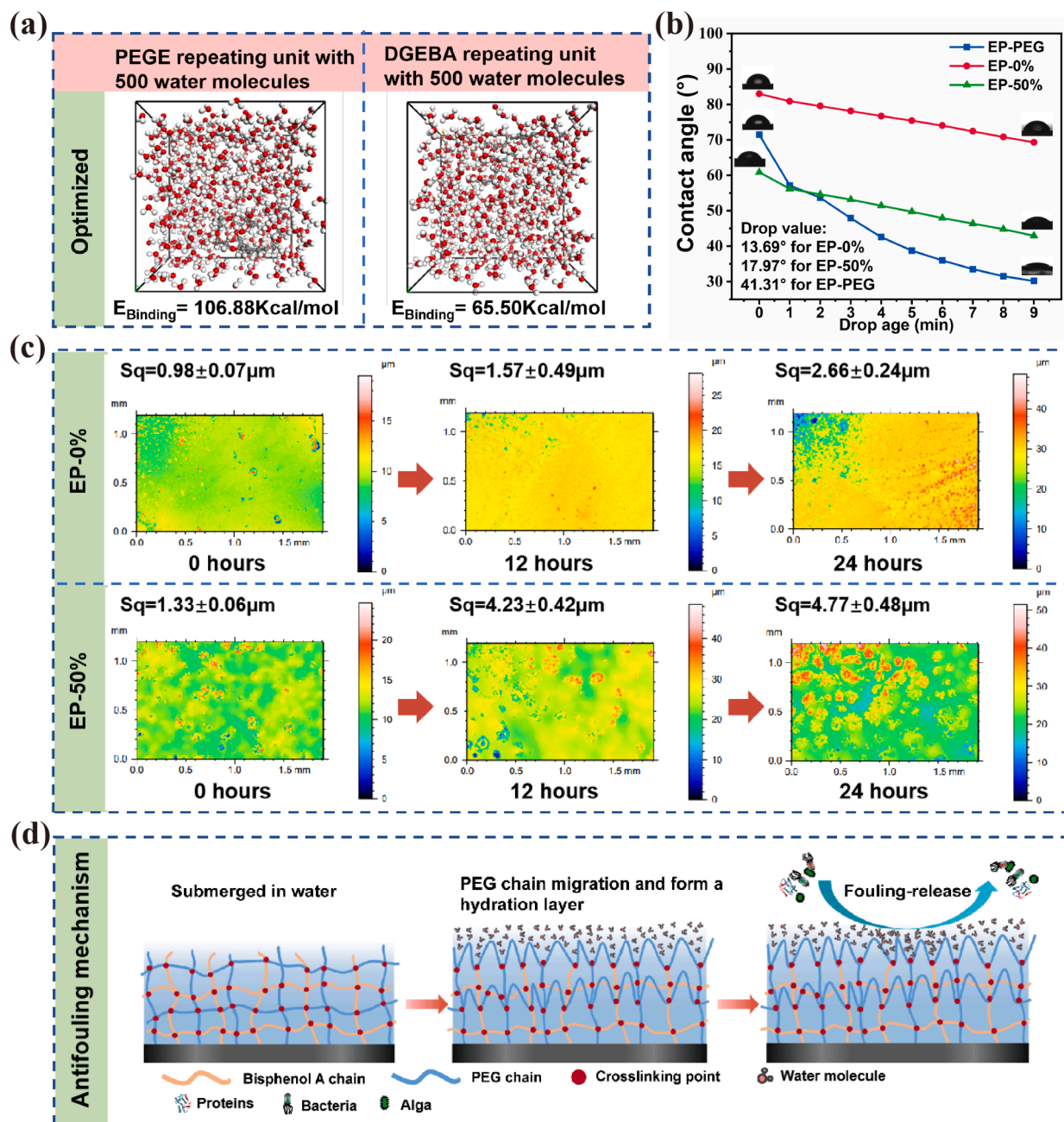


Fig. 6. (a) Snapshots of the PEGE repeating unit with 500 water molecules, DGEBA repeating unit with 500 water molecules, and their binding energy, (b) Change in contact angle of the coating over 10 min, (c) Surface roughness of EP-0 % and EP-50 % after 0 h, 12 h, and 24 h immersion in water, (d) Antifouling mechanism of cross-linked PEG chains by forming a hydration layer.

their binding energy was higher, reaching 106.88 Kcal/mol, which was higher than that of the DGEBA repeating units, which was 65.50 Kcal/mol. Due to the strong affinity of PEG chains for water, in water, the PEG chains migrate towards the surface on a micron scale, making the coating more hydrophilic. The change of contact angle of the coatings in ten minutes is shown in Fig. 6b. The initial contact angles of EP-0 % and EP-50 % were 82.98 and 60.88 degrees, respectively, and decreased by 13.69 and 17.79 degrees after ten minutes due to the difference of PEG content. The contact angle of EP-PEG without DGEBA even decreased by 41.31 degrees in the same period. The migration of PEG in water to the outside made the coating more hydrophilic. We immersed EP-0 % and EP-50 % in artificial seawater and studied the changes in surface roughness. The initial surface roughness of EP-0 % and EP-50 %, and the surface roughness after 12 and 24 h of water are shown in Fig. 6c, and

the three-dimensional surface profiles are shown in Fig. S7. Both coatings have micrometer-scale roughness, and the surface roughness increases to different degrees after immersion in artificial seawater. The high initial surface roughness of EP-50 %, which increases even more after immersion, is due to the migration of the more hydrophilic PEG chains to the water interface and facilitates the formation of a hydration layer to improve the antifouling properties of the coating [49]. The antifouling process of crosslinked PEG is summarized as shown in Fig. 6d. When the coating is placed in an aqueous environment, the greater affinity of the PEG chain for water makes the PEG chains migrate to the surface of the coating, and the conformation of the PEG chains changes to a helical structure, which reaches equilibrium driven by osmotic pressure and constraints at the cross-linking points at both ends. The PEG chains that appear on the surface of the material can resist the

adhesion of dirt by forming a hydrated layer that acts as a thermodynamic barrier [50,51].

The evidence above showed that the in-situ generation of multiple β -amino alcohol structures gives the resin a potent antimicrobial effect. However, the biocidal mechanism of the resin is still unclear. To understand the reason for the different antimicrobial capabilities brought about by using DAP and DAHP as curing agents, we analyzed the charged states of the two characteristic structures to explain the differences in their antimicrobial capabilities. Therefore, we chose the characteristic structures in these two epoxy resins named DAP-Derivative and DAHP-Derivative, respectively, and the schematic diagrams of these two characteristic structures are shown in Fig. 7a. Material Studio software calculates the atomic charge and electrostatic potential diagrams of DAP-Derivative and DAHP-Derivative and the results of the calculations are shown in Fig. 7a. There is a difference between the charge distributions of the DAP-Derivative and the DAHP-Derivative. The central carbon atom of DAHP-Derivative shows an atomic charge

of 0.217 Hartree, while the central carbon atom of DAP-Derivative has an atomic charge of -0.107 Hartree. The atomic charge calculation shows that the carbon atom connected to the β -amino alcohols in DAHP-Derivative has an intense positive charge center, and this conclusion can also be verified again by the electrostatic potential diagram. The vicinity of the β -amino alcohol of DAHP-Derivative shows a higher electrostatic potential. The reason for this phenomenon is that, on the one hand, the carbon atom is directly connected to the hydroxyl group. On the other hand, there are two nitrogen atoms in the β -position, and the induced effect of these two structures leads to a shift of the electron cloud in the vicinity of the carbon atom. As a result, the carbon atom shows a significant positive charge.

In general, positively charged molecules have a significant bactericidal ability due to their ability to attract and penetrate the negatively charged phospholipid bilayer in the bacterial cell membrane or by disrupting the charge balance of the transmembrane potential [52]. The interaction of positively charged structures with the phospholipid

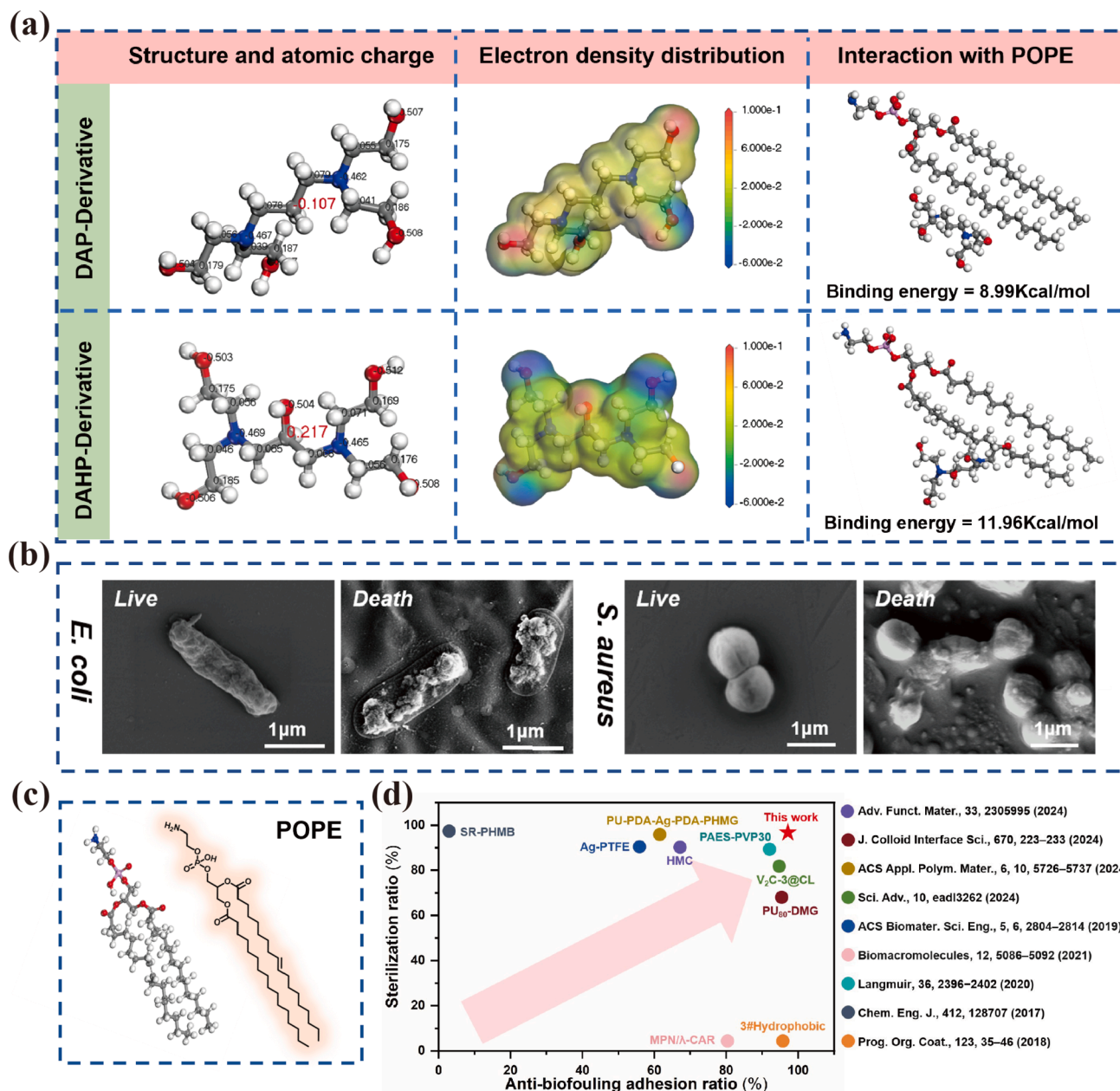


Fig. 7. (a) Chemical structures, charge properties, and interaction with POPE of DAHP-Derivative and DAP-Derivative, (b) SEM images of *E. coli* and *S. aureus*, (c) Chemical structure of POPE, and (d) Comparison of sterilizing rate and anti-biofouling rate of this work with literature.

bilayer in the bacterial cell membrane affects the bactericidal ability of the compounds [53]. To deeply verify the difference between the antimicrobial capacity of DAHP-Derivative and DAP-Derivative structures, we calculated the binding energy between these two characteristic structures and bacterial phospholipid molecules to simulate the interaction of these two structures on bacterial cell membranes. Phospholipid 1-palmitoyl-2-oleoyl-*sn*-glycerol-3-phosphoethanolamine (POPE) is one of the most representative phospholipid components of the bacterial outer membrane, accounting for 75 % of the total phospholipid composition in *E. coli* [54]. The structural formula of POPE is shown in Fig. 7c. The interaction between the designed resin and the bacterial cell membrane was simulated by calculating the magnitude of DAHP-Derivative and DAP-Derivative binding energy to POPE.

The related calculation results are also shown in Fig. 7a. DAHP-Derivative has a higher affinity for POPE. The binding energy of the DAHP-Derivative-POPE system is 11.96 kcal/mol. In comparison, the binding energy of the DAP-Derivative-POPE system is 8.99 kcal/mol, which proves that the binding between DAHP-Derivative and POPE is stronger. The greater binding energy between DAHP-Derivative and POPE proves that DAHP-Derivative can interact more strongly with the bacterial phospholipid bilayer to imbalance the bacterial cell membrane potential, thus leading to bacterial death [55]. To verify this conjecture, the morphology of *E. coli* and *S. aureus* adhered to EP-0 % was observed by SEM. As shown in Fig. 7b, compared to live bacteria, the cell membranes of the dead *E. coli* and *S. aureus* on the surface of EP-0 % were ruptured, and the intracellular material was outflowed. Considering the above calculations and experimental evidence, this characteristic structure of DAHP-Derivative contains positively charged centers that can strongly interact with the phospholipid bilayer in the cell membrane and affect the normal functioning of the cell membrane. The result is the failure of the cell membrane and the death of bacteria due to the overflow of intracellular material. This characteristic bactericidal structure combined with the antifouling effect of PEG, the material exhibits greater antifouling advantages than other materials (Fig. 7d), promising longer-lasting marine antifouling.

4. Conclusions

In summary, the epoxy systems comprising DGEBA and PEGE, which inherently generate a β -amino alcohol structure, have demonstrated outstanding bactericidal characteristics, resistance to fouling adherence, and favorable mechanical qualities. The design of the DAHP-Derivative structure paves the way for novel antimicrobial epoxy resins that can be produced on-site using β -amino alcohols as the curing agent. Calculations performed using Material Studio software indicate that the DAHP-Derivative structure, the central carbon atom in the derivative structure, exhibits a significant positive charge. This positive charge enables a more robust interaction with the phospholipid bilayer of the bacteria, resulting in the disruption of the bacterial cell membrane and subsequent bacterial death. The introduced PEG chains can generate a hydration layer by binding water molecules through hydrogen bonding. This results in the coating having exceptional resistance to adhesion by protein, bacteria, algal, and other marine fouling organisms. Epoxy resins possessing a DAHP-Derivative structure exhibit a bactericidal rate of 100 %. Epoxy resins with over 30 % PEG chains exhibit exceptional resistance to protein, bacterial, and algal adherence and outstanding coating characteristics. This study develops a marine antifouling coating that possesses both bactericidal properties and resistance to fouling adhesion, as well as provides a valuable method for the prevention of biofouling on sea-related equipment in challenging marine environments.

CRediT authorship contribution statement

Peizhan Cao: Writing – original draft, Data curation. Shu Tian: Methodology, Investigation. Jinyue Dai: Writing – review & editing,

Validation, Formal analysis. Jiayin Li: Formal analysis, Data curation. Guangming Lu: Validation. Xiaoqing Liu: Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2024.159057>.

Data availability

Data will be made available on request.

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